

Toxicogenomics I

Linking pollution to functional genomics

Dr. Marie Brasseur, AG Ecotoxicology

What are Toxicogenomics?

- *How environmental pollutants affect the genome of living organisms and its expression dynamics*
- Identify the molecular pathways underlying biological responses that are affected/induced by toxicant exposure
- Important to gain **mechanistic insights**

The central dogma of molecular biology



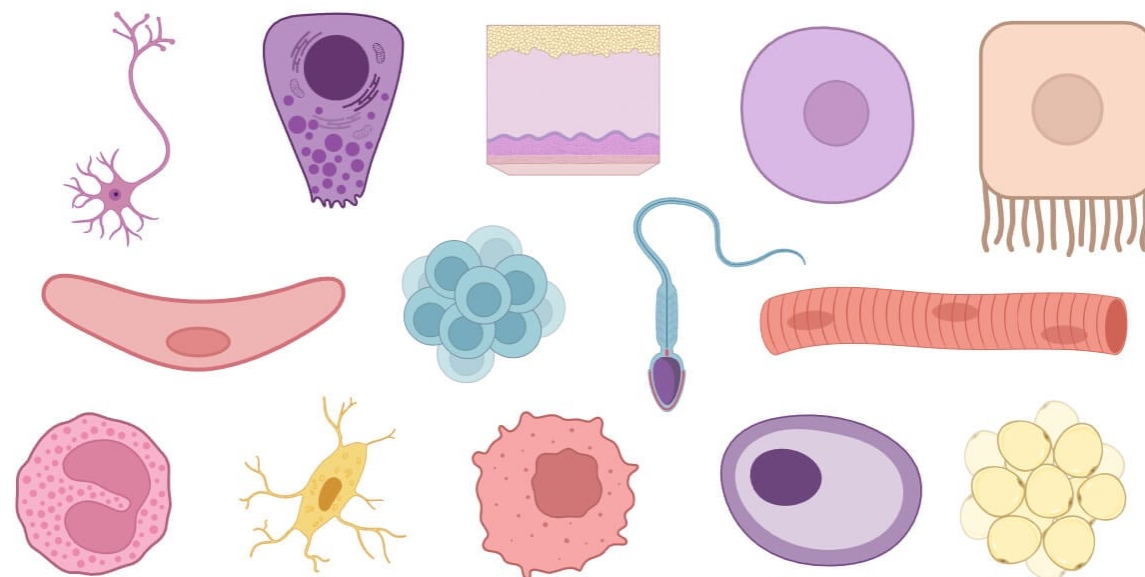
Different cell types have the same genotype...

Genotype

- Encodes the genetic information
- Static

Phenotype

- Expression of genetic information
- Plastic



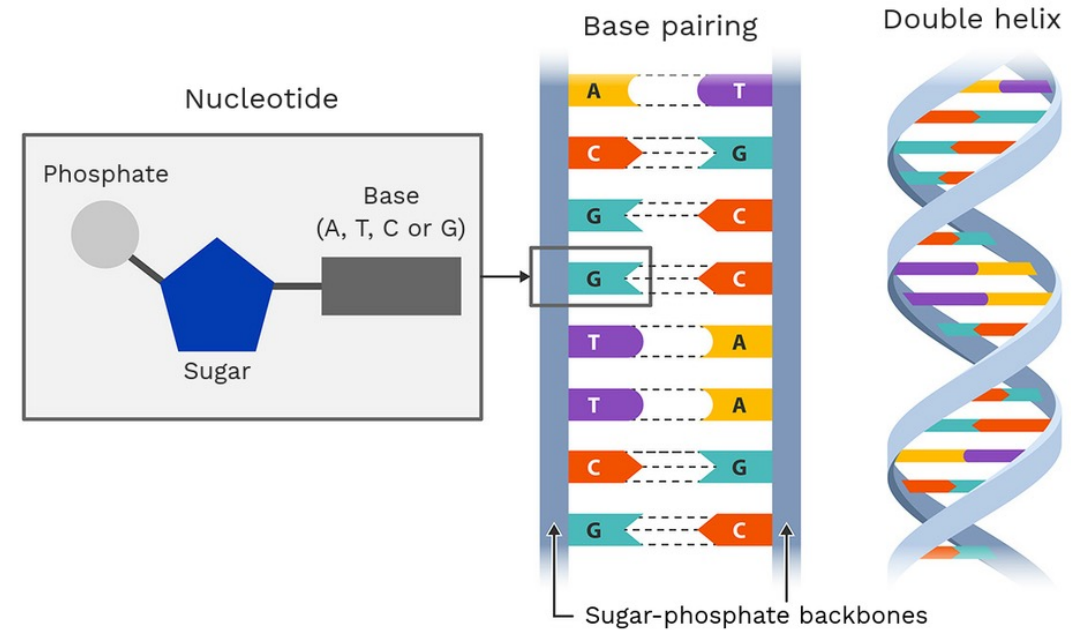
... but a very different phenotype

The central dogma of molecular biology



Encoding of genetic information:

- Sequence of bases
- Complementary base pairing (A-T & C-G)

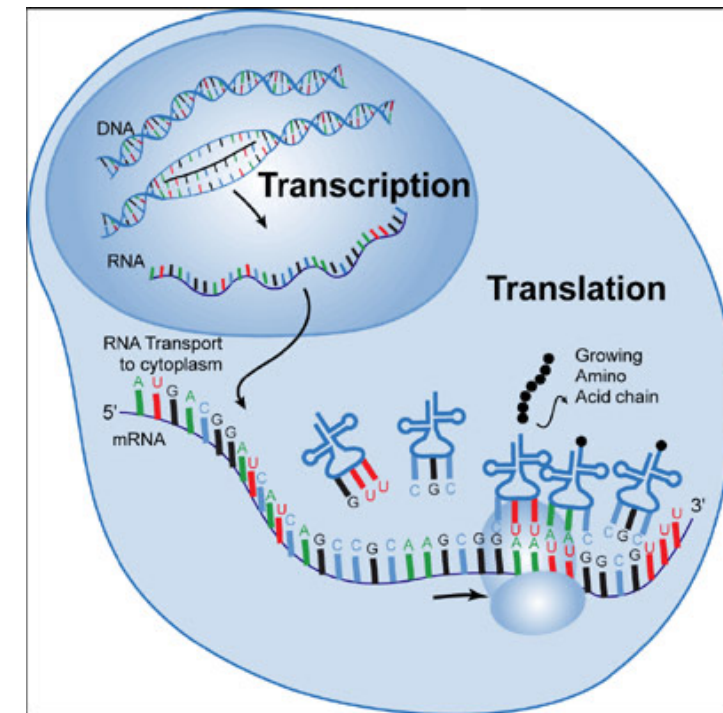


The central dogma of molecular biology



Encoding of genetic information:

- Sequence of bases
- Complementary base pairing (A-T & C-G)
- Transcription: DNA (double stranded) -> RNA (single stranded) via complementary base pairing

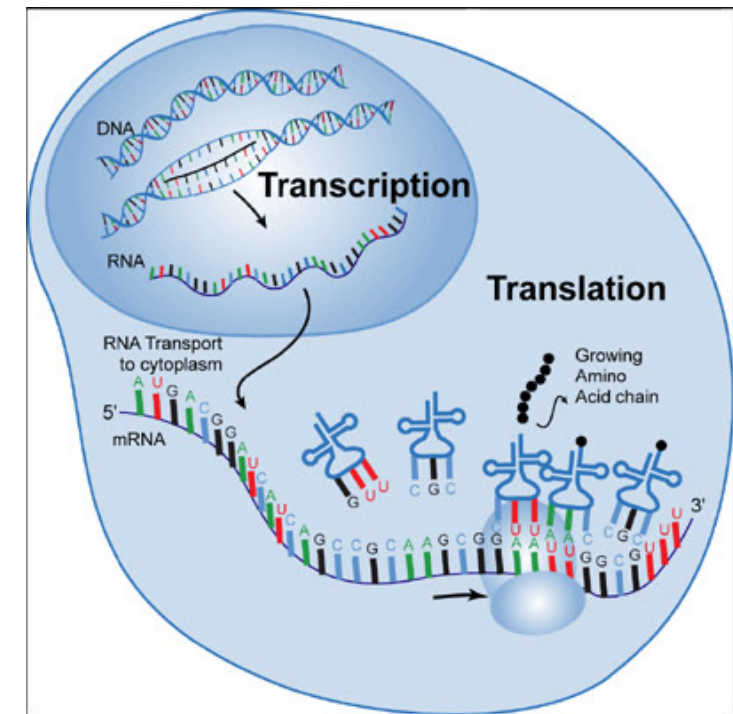


The central dogma of molecular biology



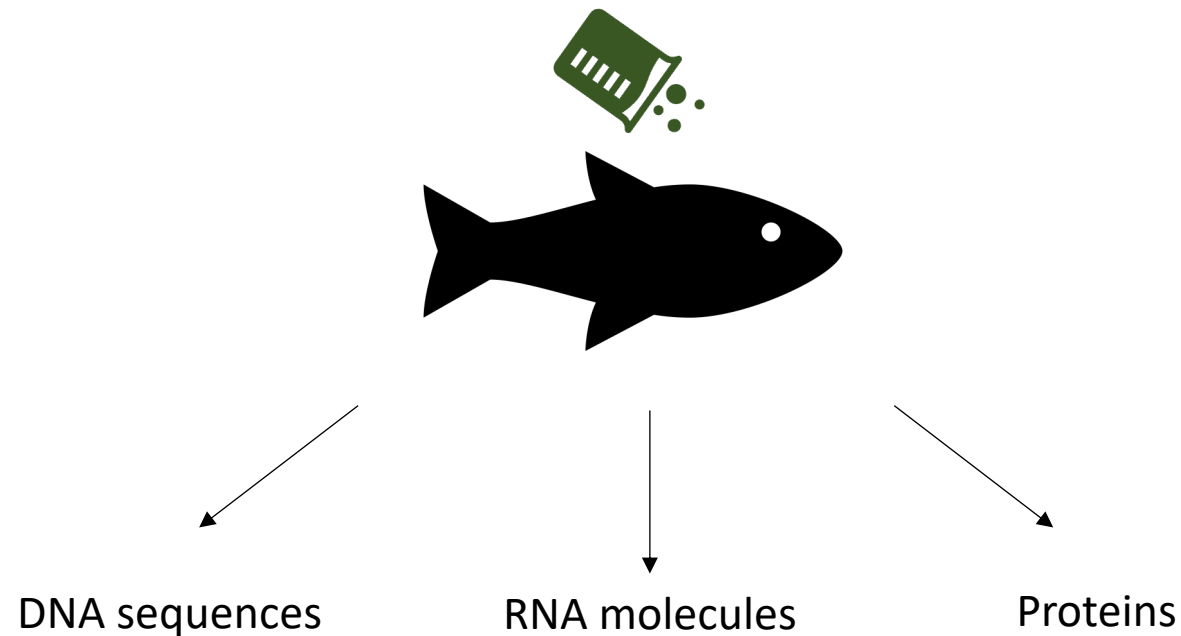
Encoding of genetic information:

- Sequence of bases
- Complementary base pairing (A-T & C-G)
- Transcription: DNA (double stranded) -> RNA (single stranded) via complementary base pairing
- Translation: RNA triplet -> amino acid



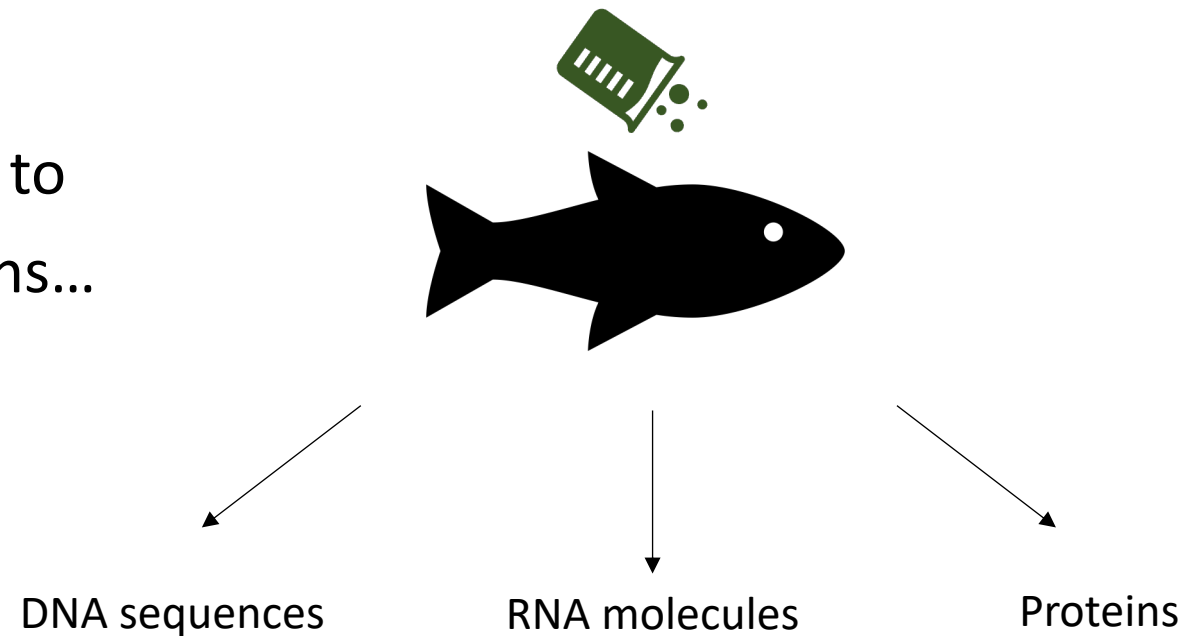
What are Toxicogenomics?

- *How environmental pollutants affect the genome of living organisms and its expression dynamics*



What are Toxicogenomics?

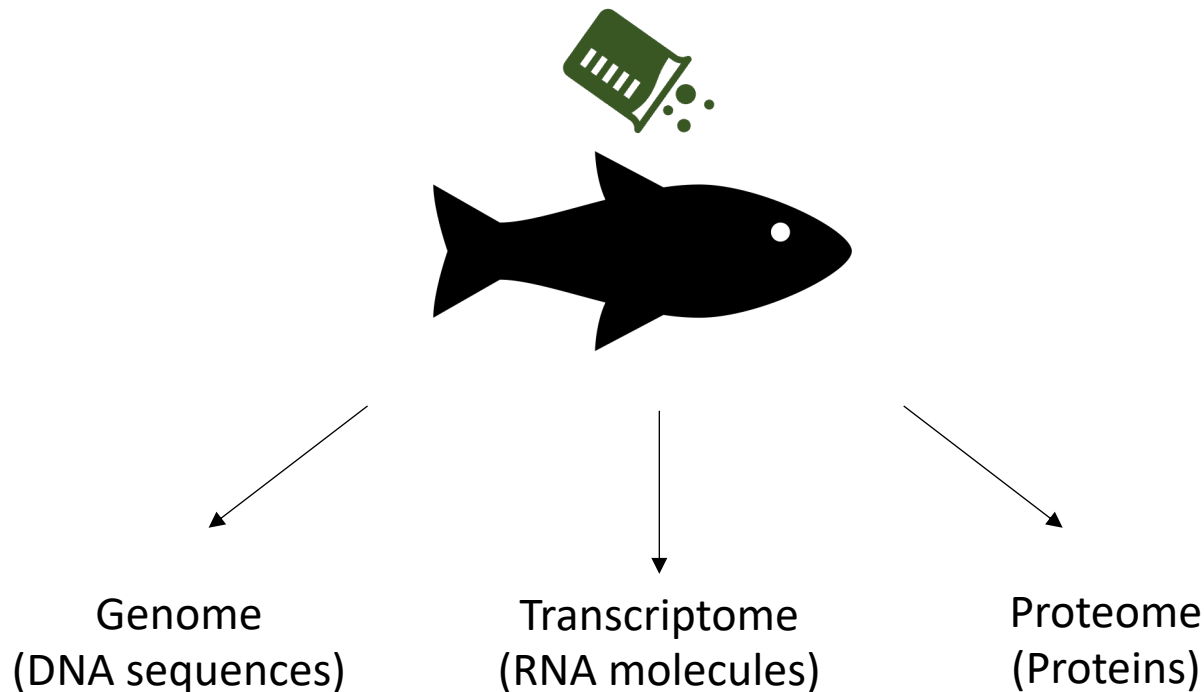
- *How environmental pollutants affect the genome of living organisms and its expression dynamics*
- While we were earlier limited to single genes, transcripts, proteins...



What are Toxicogenomics?

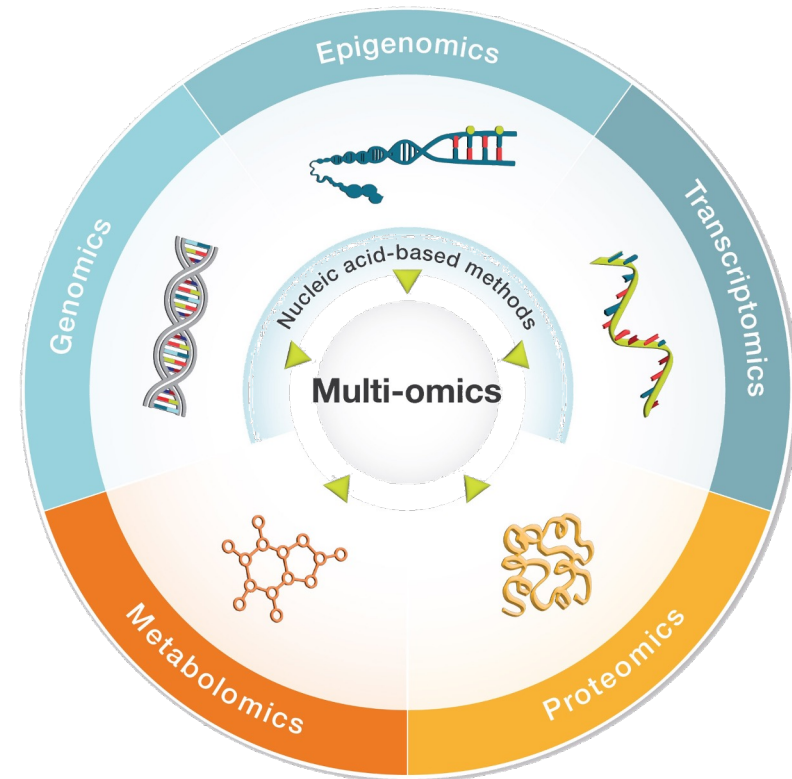
- *How environmental pollutants affect the genome of living organisms and its expression dynamics*

...we can nowadays assess whole genomes, transcriptomes, proteomes (metabolomes, epigenomes, lipidomes...)



What are Toxicogenomics?

- *How environmental pollutants affect the (epi)genome, the transcriptome, the proteome and the metabolome of living organisms*
- Integration of **–OMICS** in ecotoxicology



Recap questions

- What is stated by the central dogma of molecular biology?
- How is genetic information encoded?
- How can we infer the (unknown) RNA sequence from a (known) DNA sequence?
- What is the difference between the genome and the transcriptome?

The technical part: -OMICS

- High-throughput technologies to measure

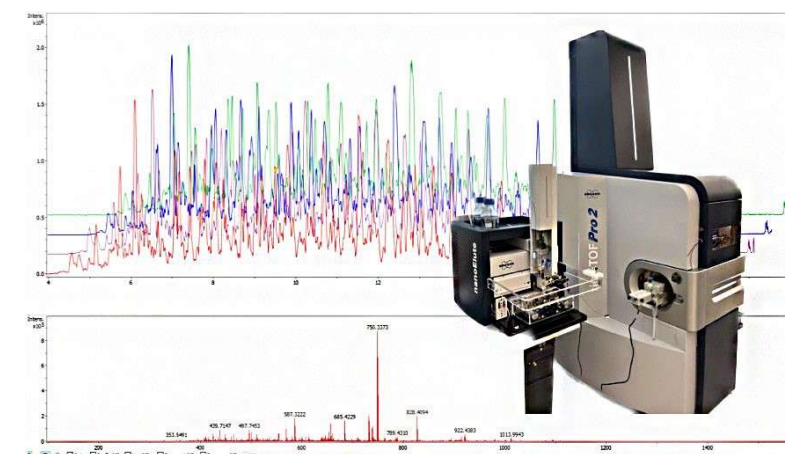
- (Epi-)Genome, Transcriptome

→ Based on next generation sequencing
(decode base sequence of DNA & RNA)

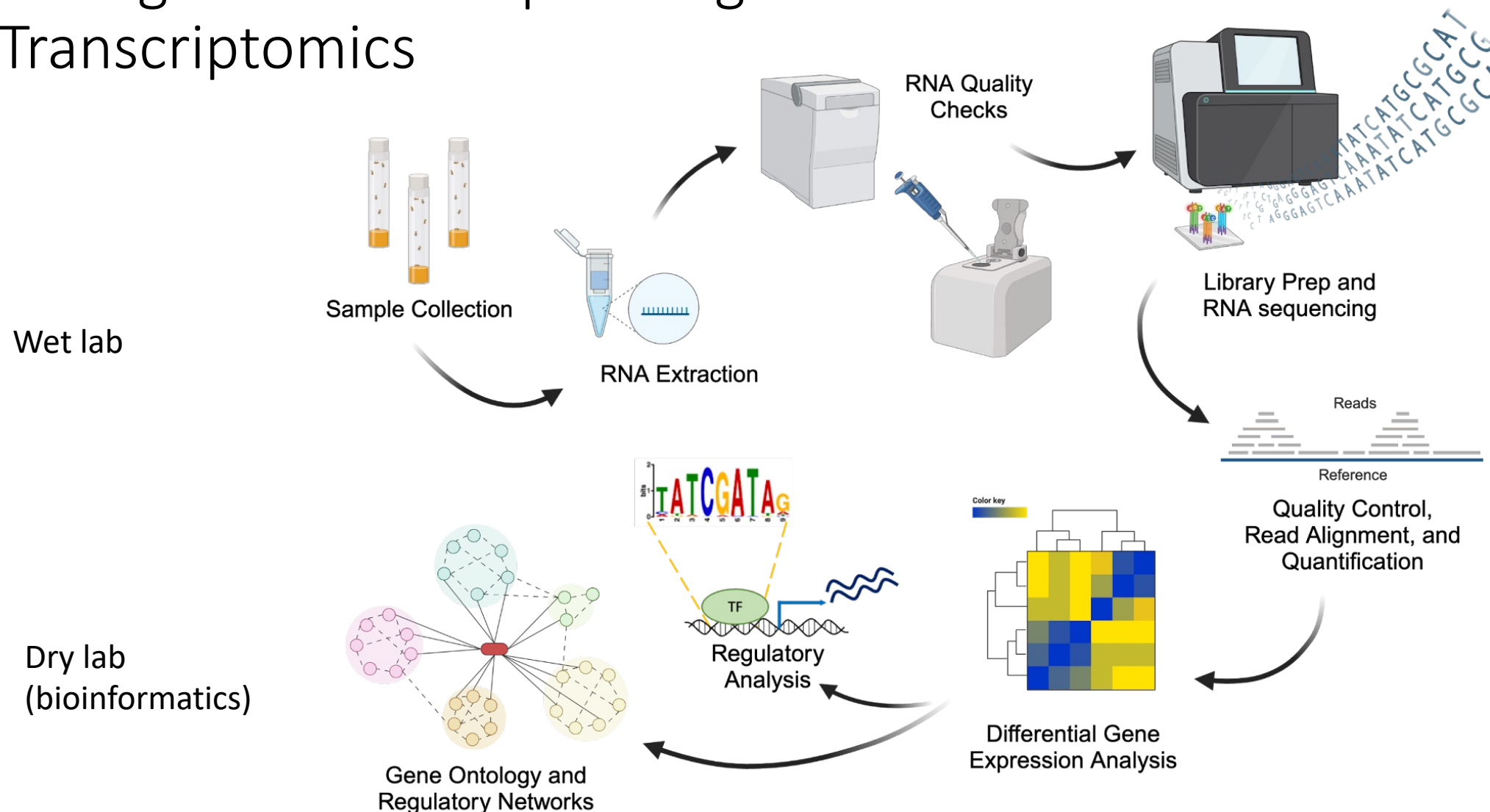


- Proteome, Metabolome, Lipidome

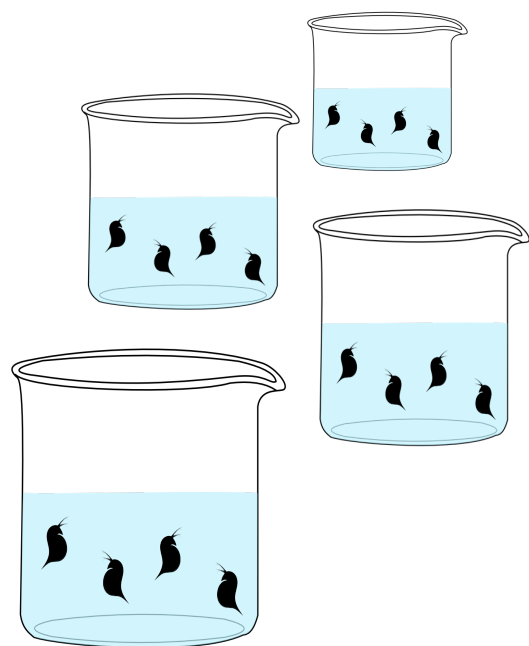
→ Based on mass spectrometry
(mass spectrum = compound specific
pattern of mass/charge ratio peaks;
not covered here)



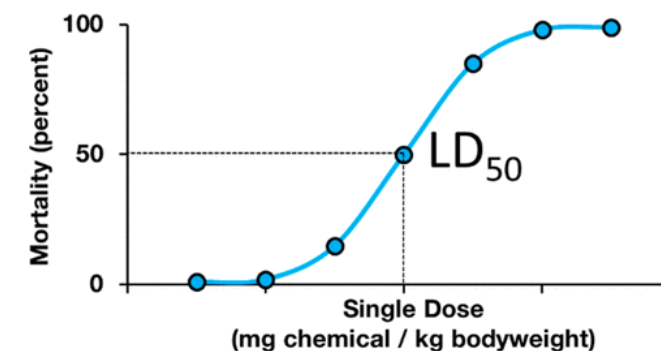
Next generation sequencing of RNA: Transcriptomics



Wet lab



Measure e.g. survival



Measure e.g. gene expression

- Pre-OMICS:
 - qPCR -> single genes
 - Microarrays -> multiple genes (hundreds to thousands)
- -OMICS
 - RNA sequencing -> global assessment of the whole transcriptome

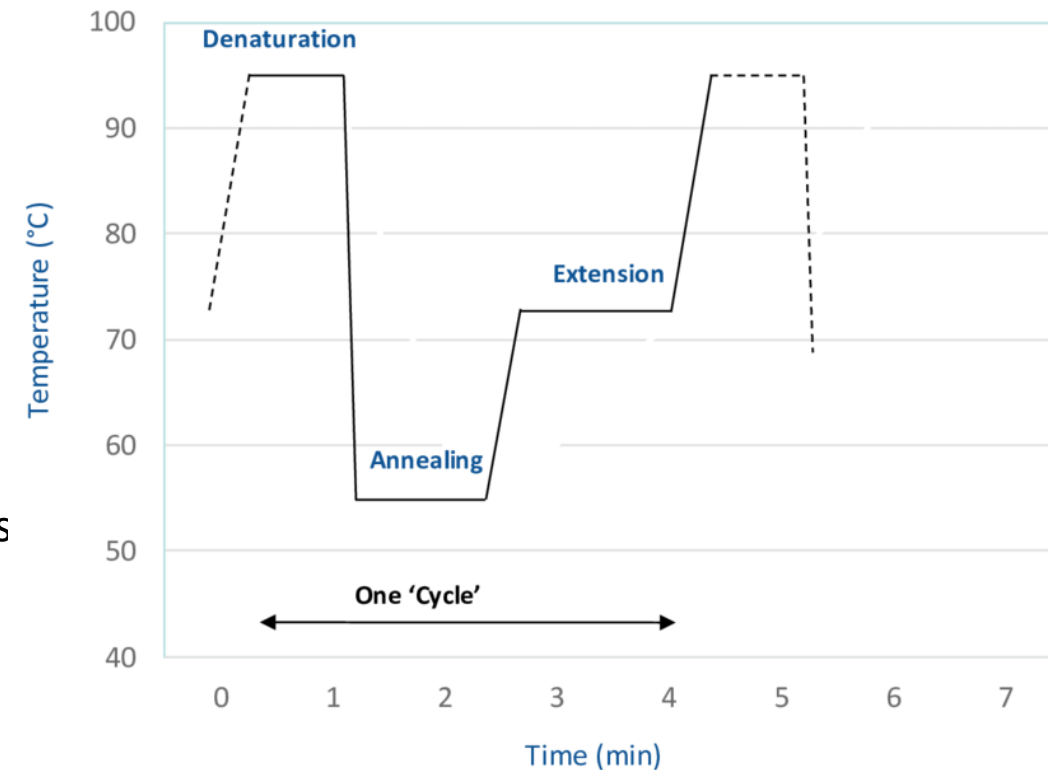
(Eco)toxicological experiment

Wet lab – (q)PCR

- PCR: Polymerase chain reaction
- Artificial DNA replication in 3 steps:
 - PCR ingredients: DNA polymerase (e.g. *Taq*), nucleotides (dNTPs), primers, template DNA

One cycle:

1. Heat denaturation: double-stranded DNA → 2 single strands
2. Primer annealing
3. Extension: DNA polymerase can bind to primer and extends the single stranded DNA (template) by synthesizing the complementary DNA strand



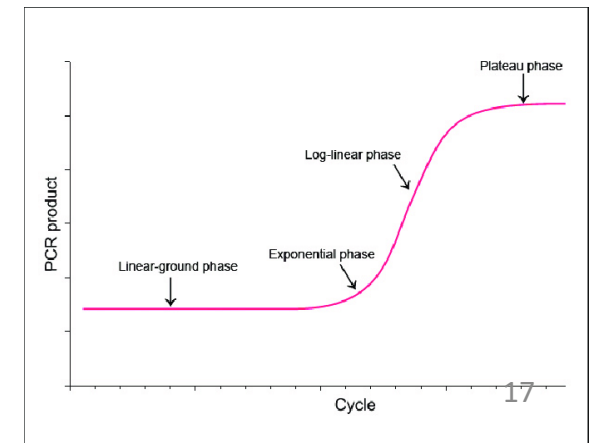
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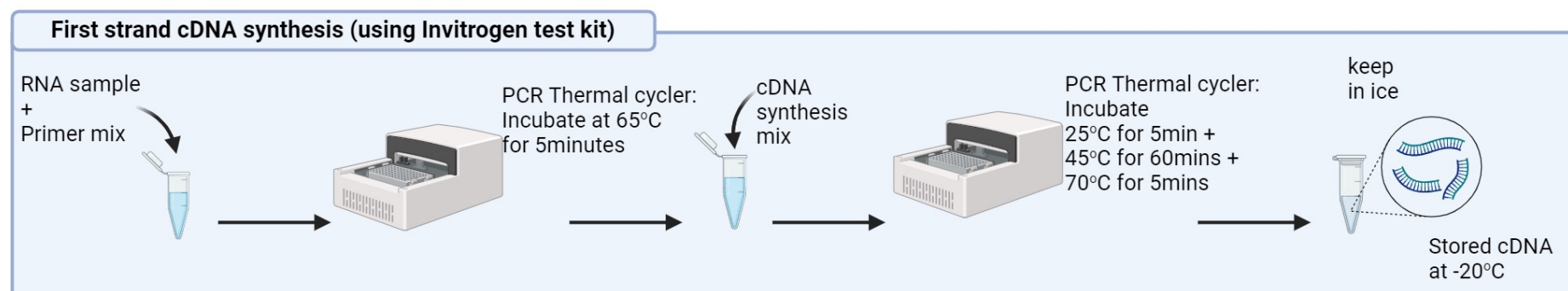
1. Heat denaturation: double-stranded DNA → 2 single strands.
2. Primer annealing
3. Extension: DNA polymerase can bind to primer and extends the single stranded DNA (template) by synthesizing the complementary DNA strand

$\sim 2^n$ copies (n =cycle number)

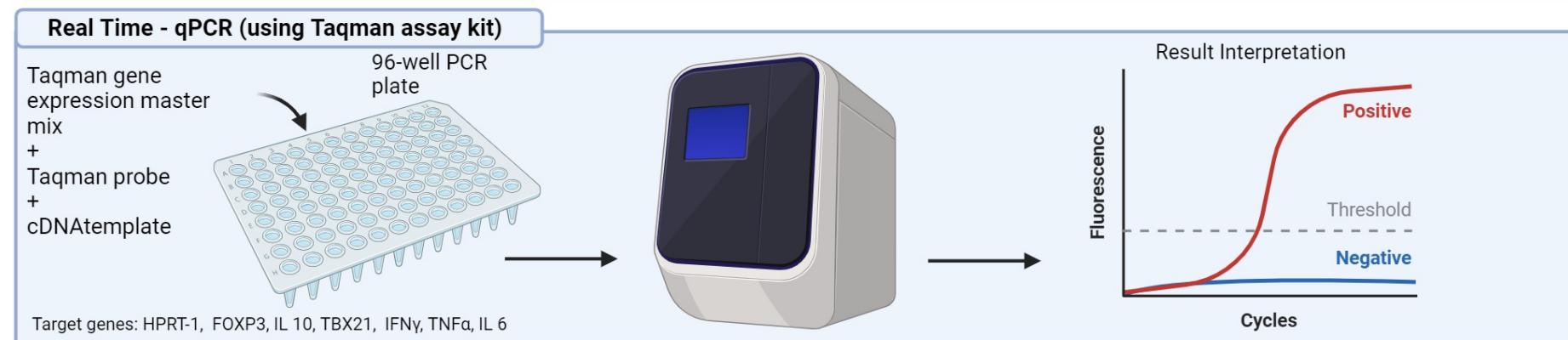


Wet lab – (q)PCR

- qPCR: quantitative Polymerase chain reaction



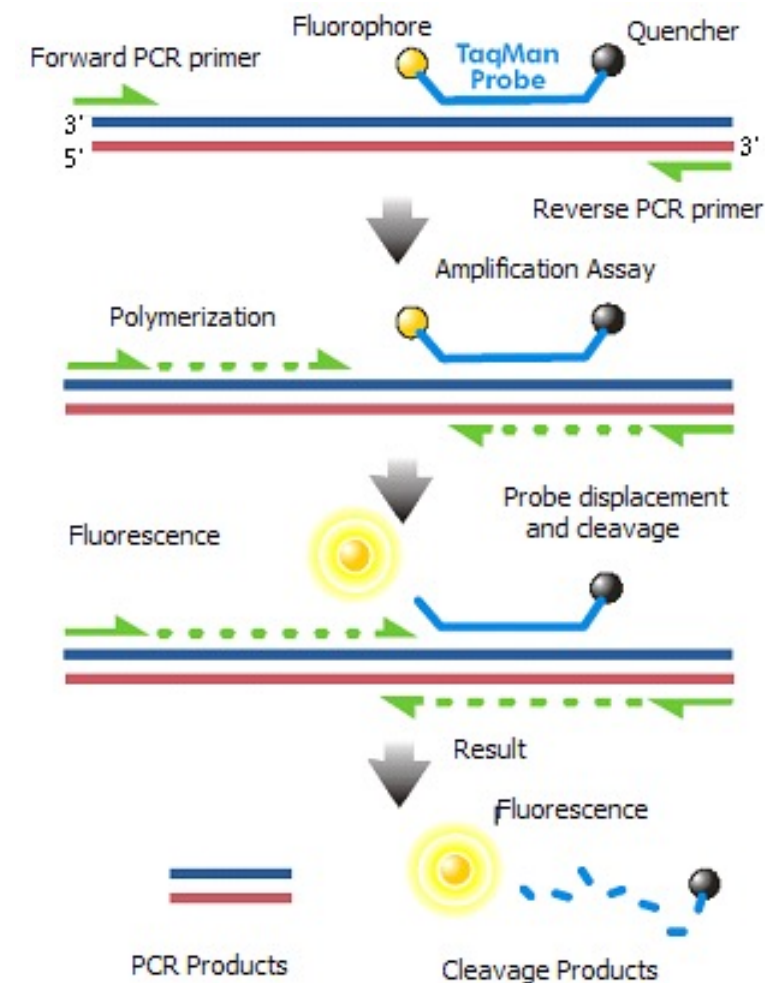
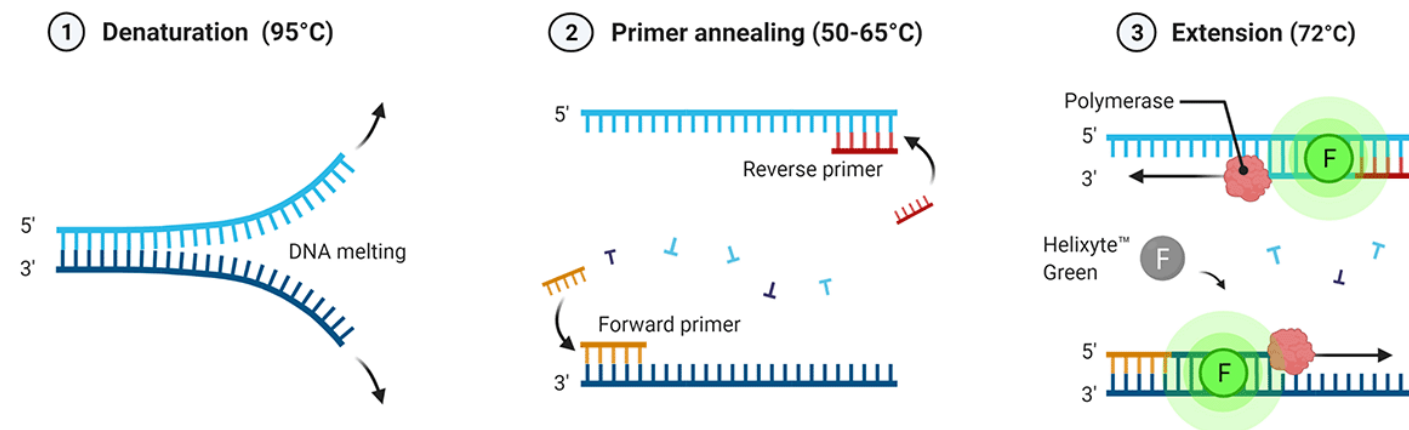
RNA must be reversely transcribed



Polymerase reaction

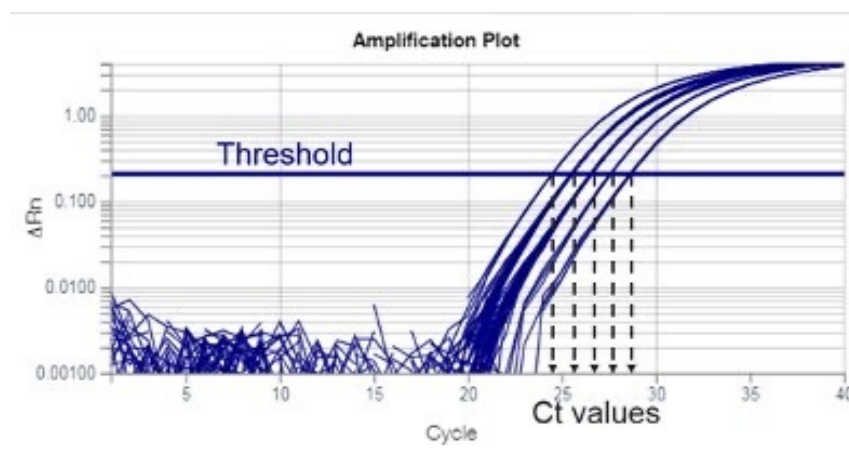
Wet lab – (q)PCR

- qPCR: quantitative Polymerase chain reaction
 - During PCR based amplification, light signal is emitted

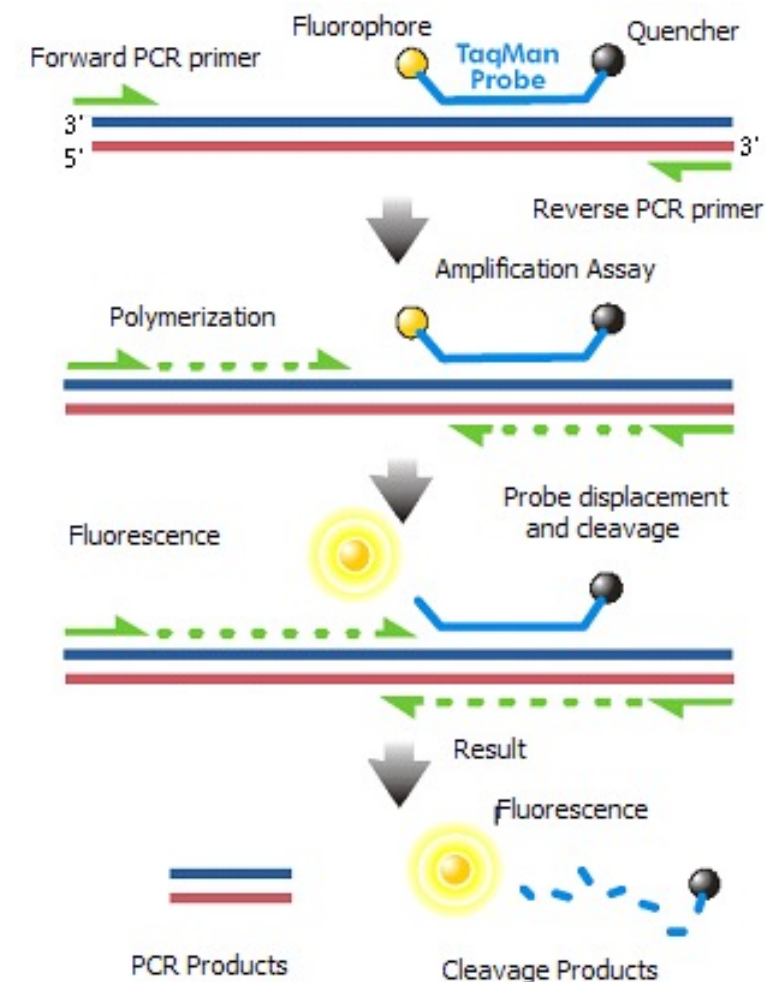


Wet lab – (q)PCR

- qPCR: quantitative Polymerase chain reaction
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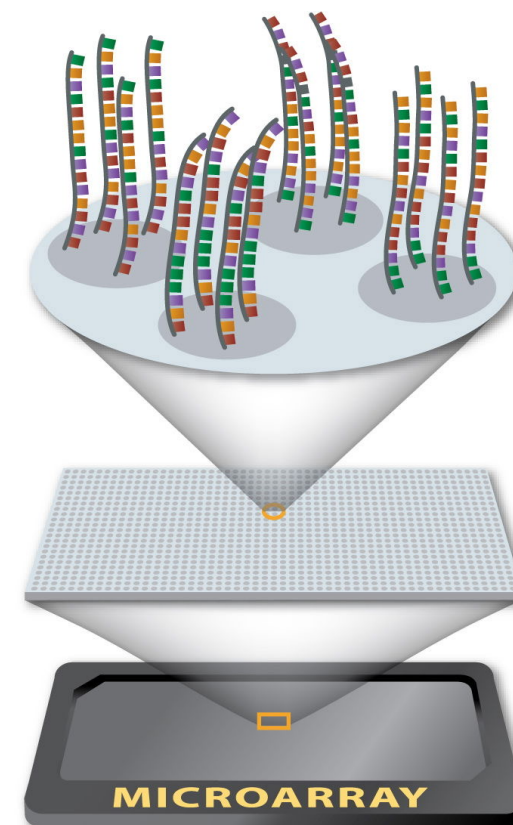
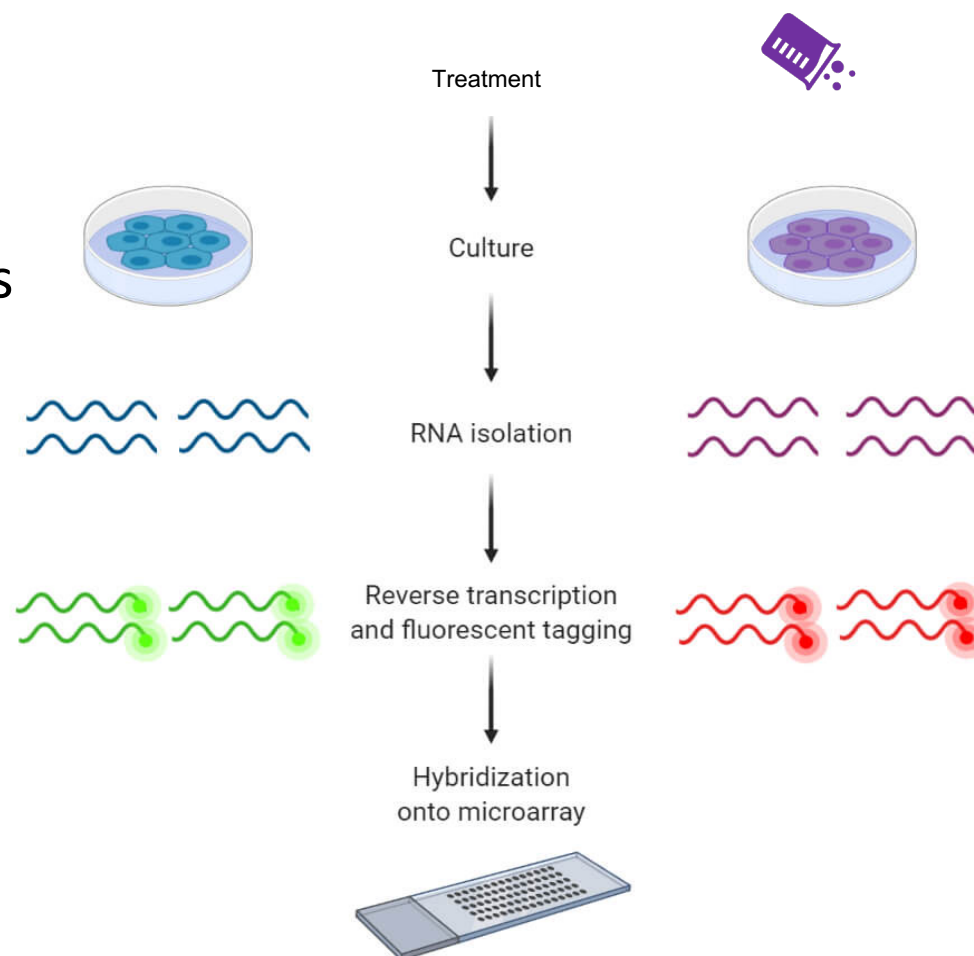
$$\text{Quantity} \sim e^{-Ct}$$



Wet lab – cDNA microarrays

Microarrays

- High throughput
(hundreds to thousands
of genes)



Wet lab – cDNA microarrays

Microarrays

- High throughput
(hundreds to thousands
of genes)
- Still prior knowledge about the
genome/transcriptome required
for probe design

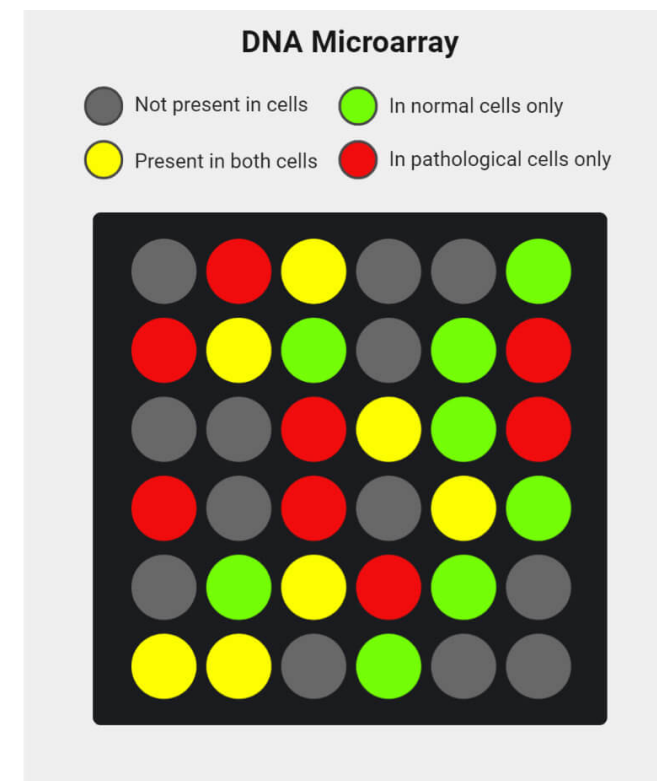
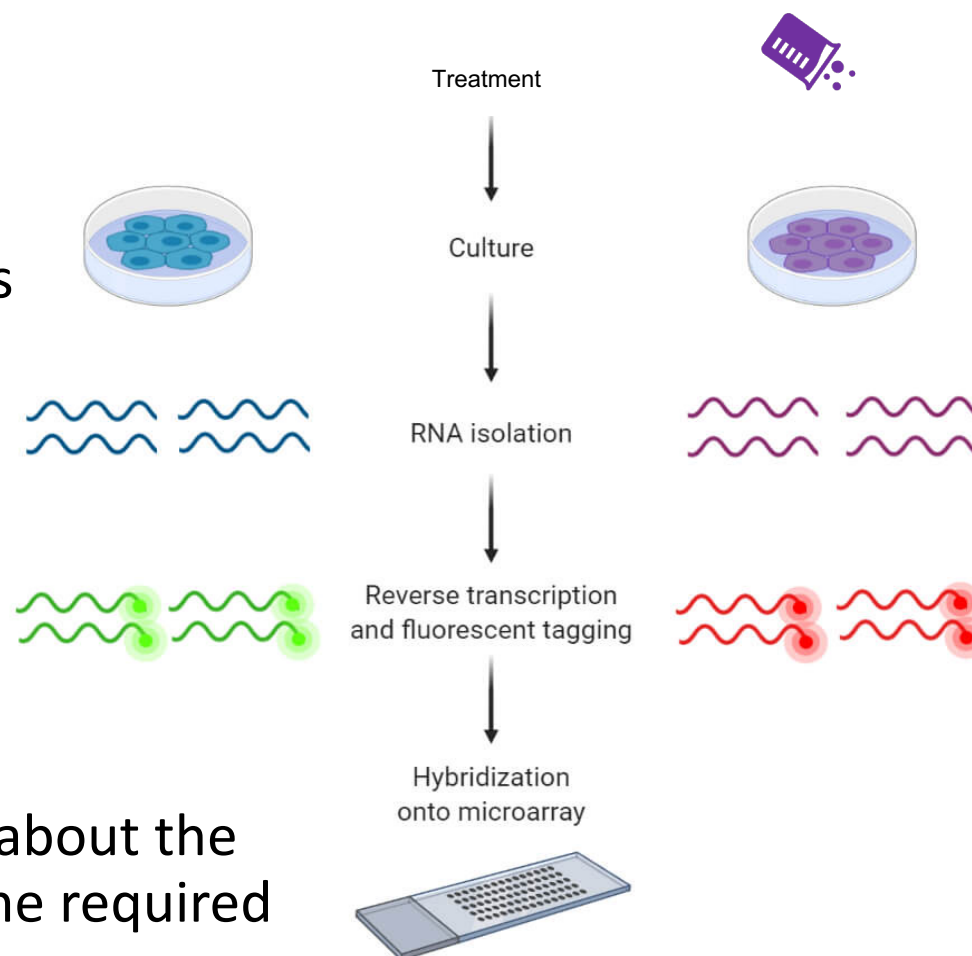
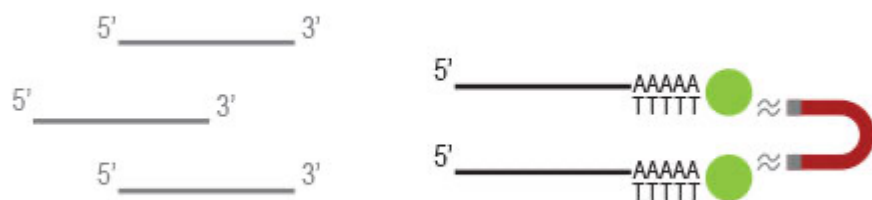
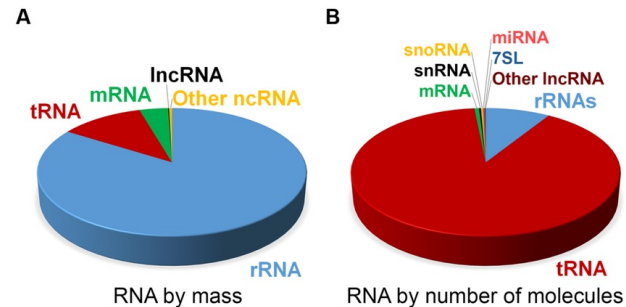


Image By Sagar Aryal, created using biorender.com

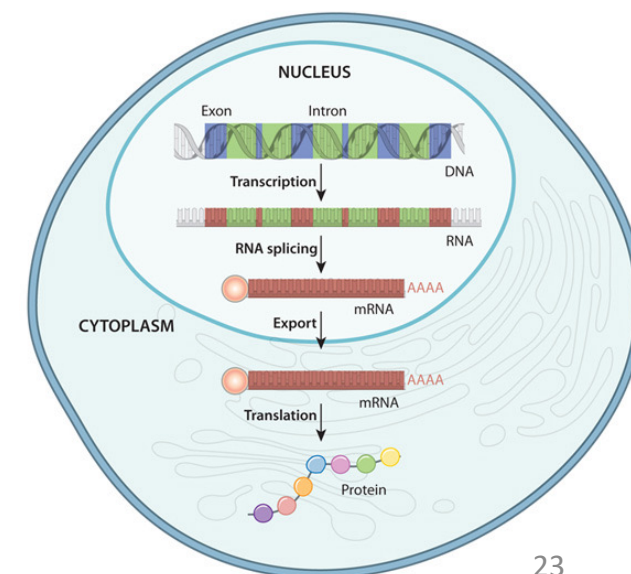
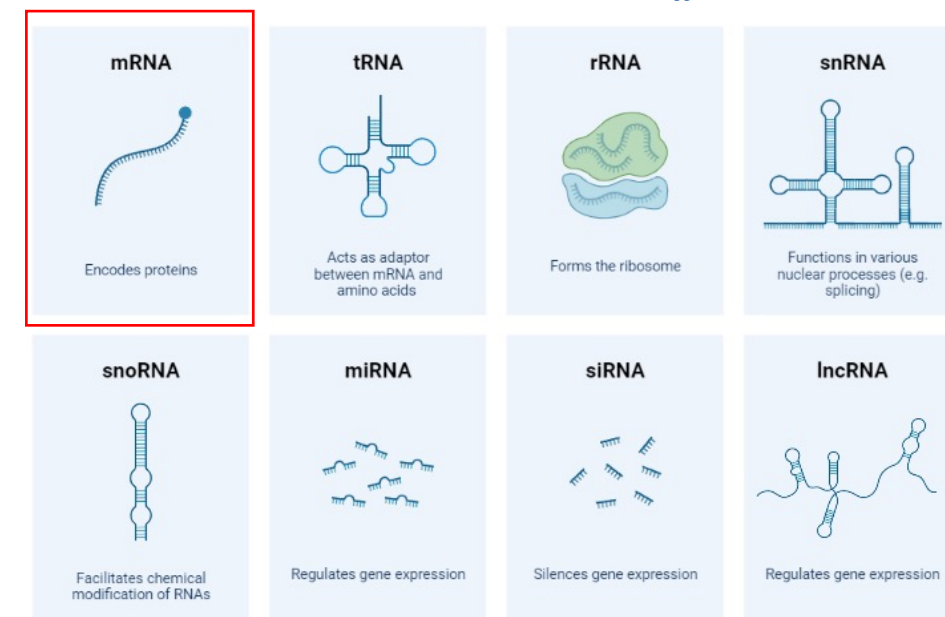
Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA



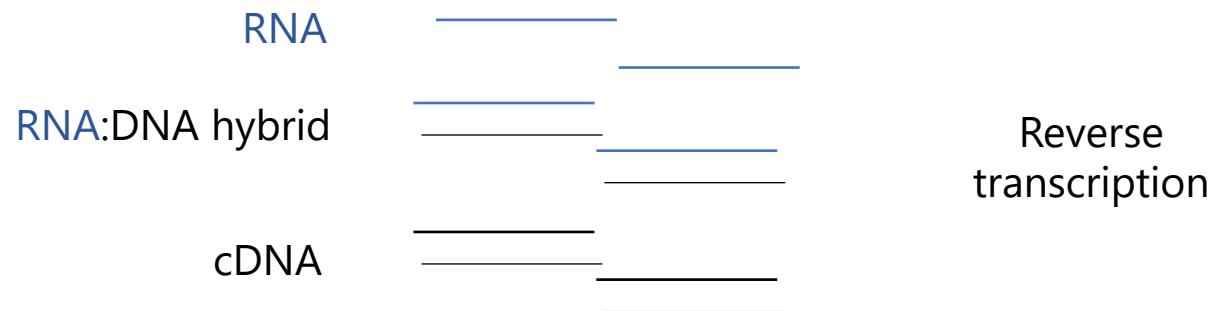
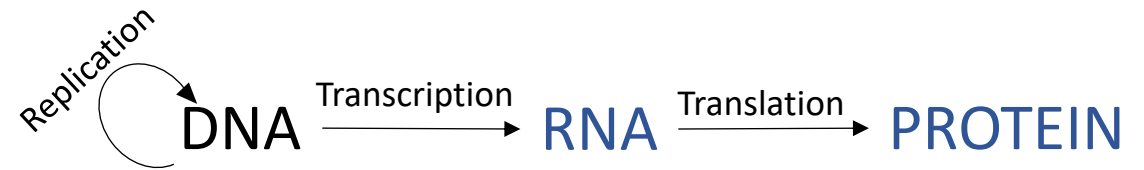
Poly(A) selection



Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
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- 2. Problem: sequencing platforms are
designed for **short DNA** fragments

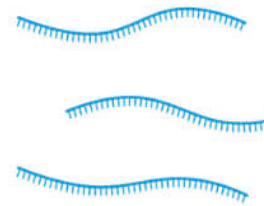


Wet lab - RNA-Seq

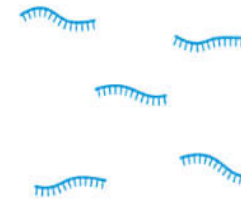
RNA sequencing = cDNA sequencing

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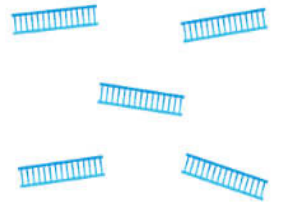
① Isolate RNA from samples



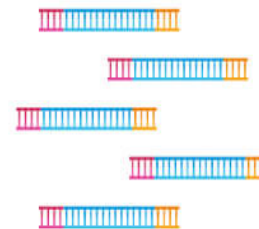
② Fragment RNA into short segments



③ Convert RNA fragments into cDNA



④ Ligate sequencing adapters and amplify



⑤ Perform NGS sequencing



Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA
- 2. Problem: sequencing platforms are designed for **short DNA** fragments

2 videos recommended:

<https://www.youtube.com/watch?v=CZeN-IgjYCo>

Or

<https://www.youtube.com/watch?v=EDVKxSNdSic>

(from Illumina; more comple)

Next Gen Sequencing

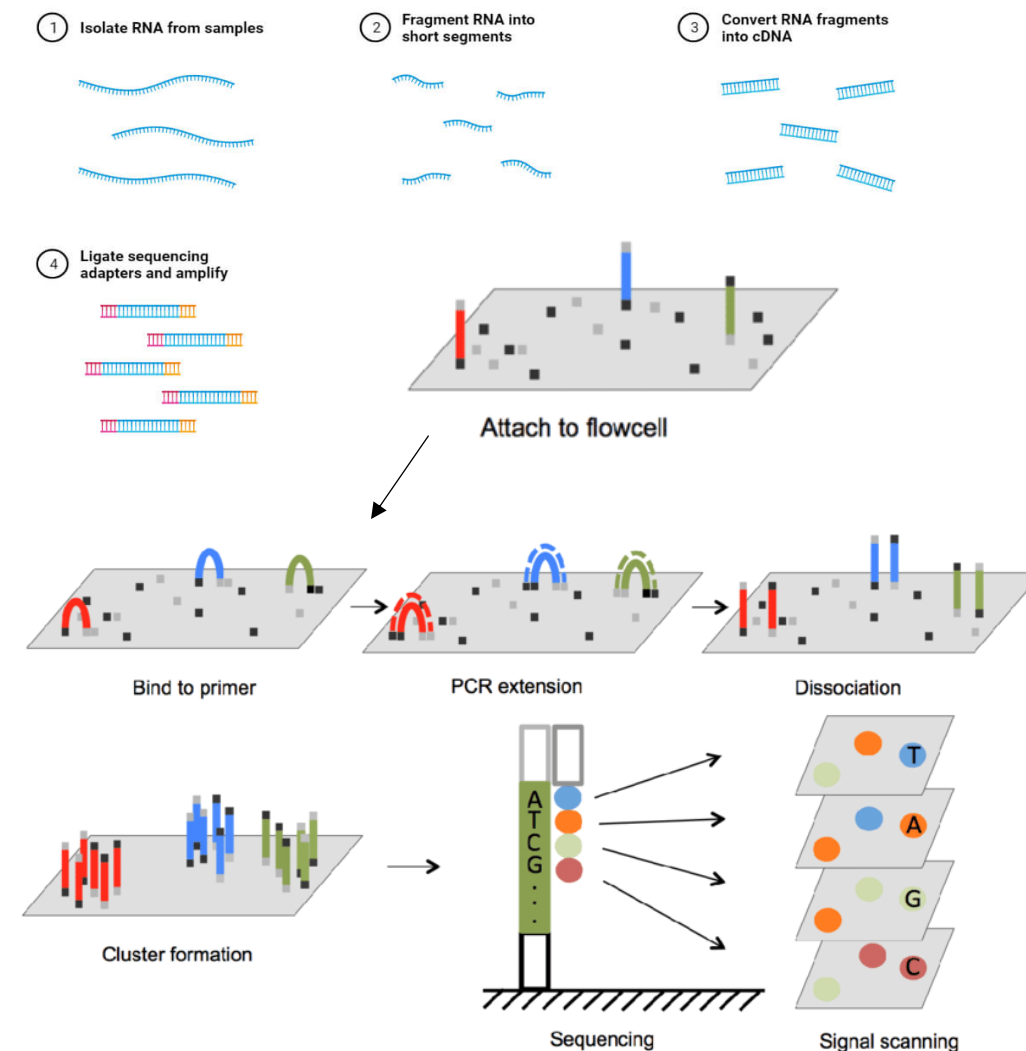


Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA
- 2. Problem: sequencing platforms are designed for **short DNA** fragments

-> Comes with new opportunities
& challenges!

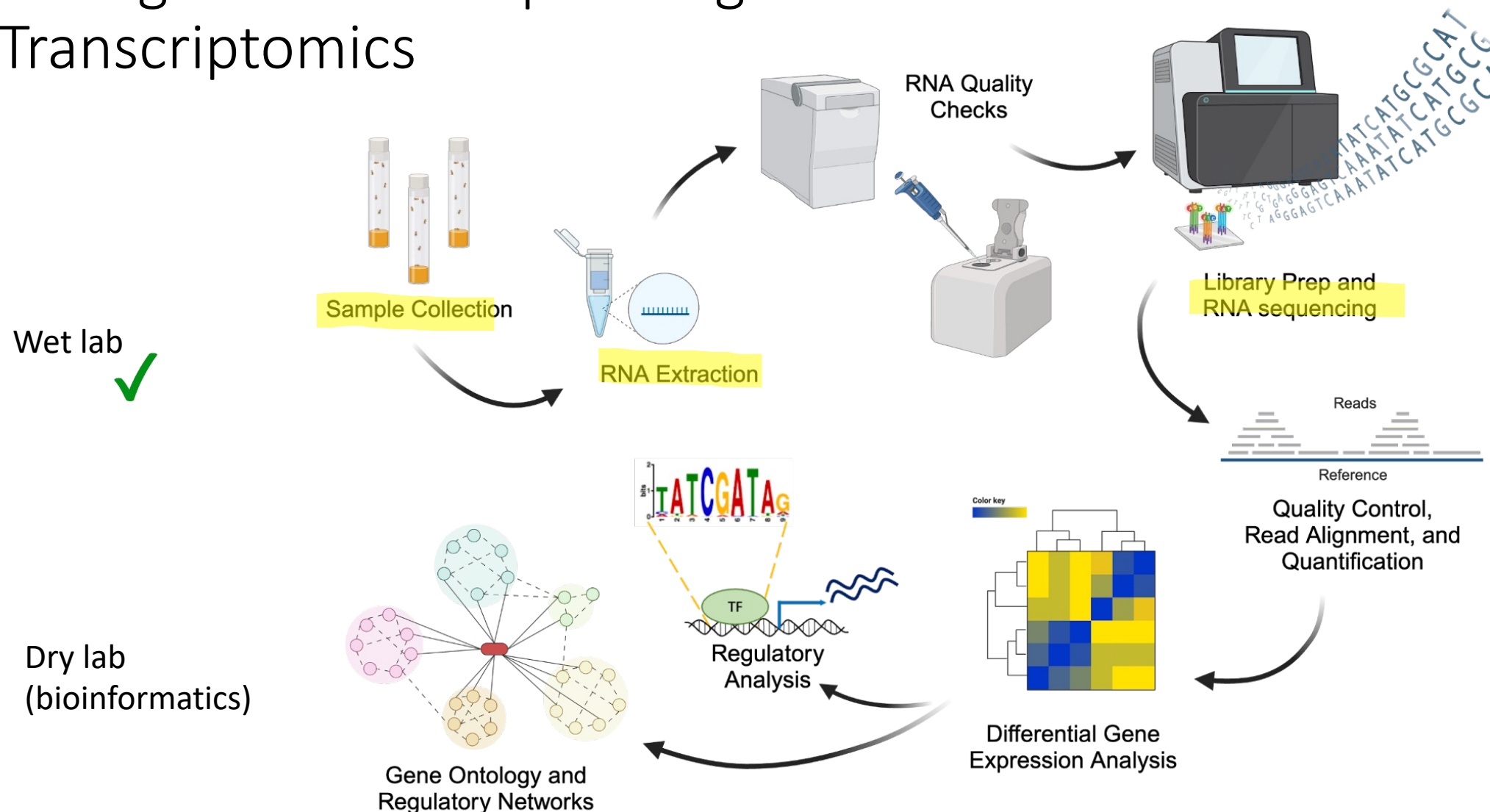


Illumina sequencing

Recap questions

- What are the three steps of a PCR?
- Why is the PCR the basis for gene expression analysis?
- How do the different gene expression methods differ (& what do they have in common) ?

Next generation sequencing of RNA: Transcriptomics



Installation

- Install these Bioconductor R packages:

```
if (!requireNamespace("BiocManager", quietly = TRUE))  
  install.packages("BiocManager")
```

```
BiocManager::install("DESeq2")
```

```
BiocManager::install("airway")
```

```
BiocManager::install("PCAtools")
```

```
BiocManager::install("apeglm")
```

- Install these standard R packages:

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
install.packages("dplyr")
```

Dry lab - Raw RNA-Seq data

@M03234:108:000000000-CFH62:1:1105:22330:22739 1:N:0:1 Header

TCTGGGTGGCCGAAAAATCAAAATAAATGTTGATATAAAACAGGGTCACCGCCTCCAGCTGGGTCAAAAAAAGTAGTATTAAATTCCTATCTGTATAAAAGCATTGTT [Read more](#)

 $+$

AFFFFFFAGFGGGGGCGGHFHFHHFHGGGHGFGGDAGHGGEGCHG2AGFHEAFFGHHEFHFHG?FGHHHHGGG@B4@F4GFHHHHHHFHDGEHHGHFHHHGGHDG3BGH
 [Quality string](#)

@M03234:108:000000000-CFH62:1:1106:13496:12956 1:N:0:1 Header

TCTGGATGTCCAAAGAACCAAAATAAATGTTGATATAATACAGGGTCGCCTCCTCCTGCAGGGTCAAAAAAGGATGTATTAAAGTTTCTATCTGTAAGTAACATTGTTA [Read more](#)

 $+$

C F F F F F F G G G G G G G G H G H H G G G G G H H H H H H H H H H H G H H H H H H G G G G H H H H H H H G H H H H H H H H H H H H H H H H H H H

Quality string

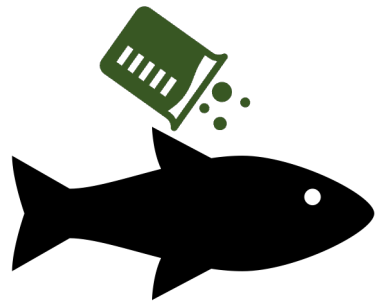
...

...

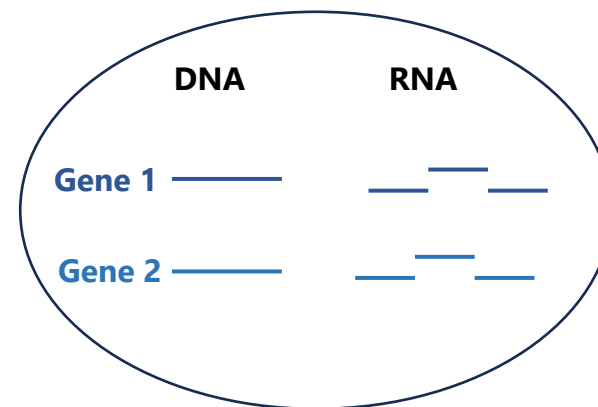
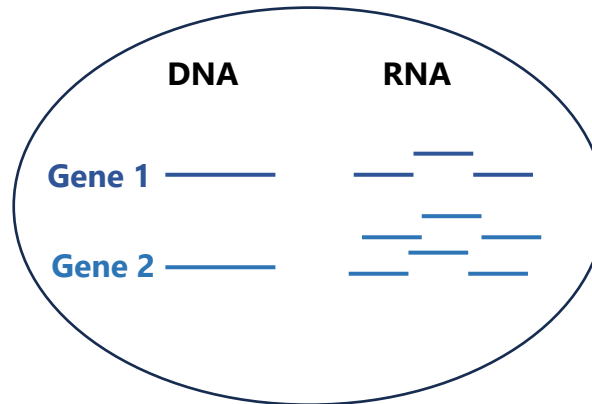
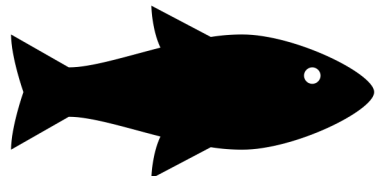
...

Remember:

The aim is to quantify gene expression changes



Vs.



- cDNA reads correspond to initial mRNA levels

BUT

- Reads (75 – 150 bp) are much shorter than the biological transcript (several kbp)

(Problem not specific for RNA seq, but the resulting challenges are greater than for DNA seq data)

Dry lab - Raw RNA-Seq data

@M03234:108:000000000-CFH62:1:1105:22330:22739 1:N:0:1 Header

TCTGGGGTGGCCGAAAAAATCAAAATAAATGTTGATATAAAACAGGGGTCACCGCCTCCAGCTGGGGTCAAAAAAAGTAGTATTAAAAATTCCTATCTGTATAAAAGCATTGTT

 $+$

AFFFFFFAGFGGGGCGGHFHFHHFHGGGHGFGGDAGHGGEGCHG2AGFHEAFFGHHEFHFHG?FGHHHHGGG@B4@F4GFHHHHHHFHDGEHHGHFHHHGGHDG3BGH [Quality string](#)

@M03234:108:000000000-CFH62:1:1106:13496:12956 1:N:0:1 [Header](#)

TCTGGATGTCCAAAGAACCAAAATAAATGTTGATATAATACAGGGTCGCCTCCTCCTGCAGGGTCAAAAAAGGATGTATTAAAGTTTCTATCTGTAAGTAACATTGTTA [Read more](#)

+

C F F F F F F G G G G G G G G H G H H G G G G G H H H H H H H H H H H G H H H H H H G G G G H H H H H H H G H H H H H H H H H H H H H H H H H H

Quality string

...

...

...

To quantify transcript abundance, millions of short reads are aligned to a reference

- Reference genome -> contains exonic and intronic elements
- Transcriptome -> contains only exonic elements

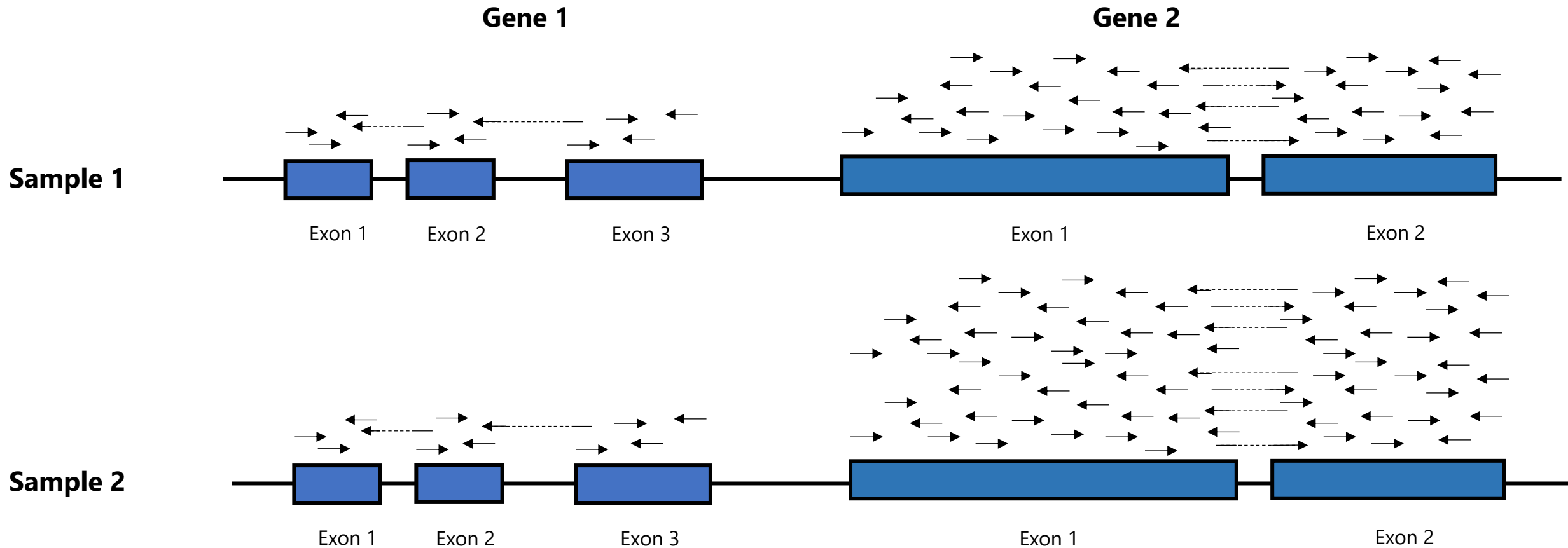
Dry lab - Reference genome

>chromosome1

GTTGCTTCTATTTAAGTTCTTTTTACTTGATATAAATATAGATAAATATATACAAAAGAAATAGTTCAGATAATCGCAAAATAATTTTCAGAGC
GAACATTTAAAATGTGTTCTGATCCAAGTTTCACTGTTACTCCTCCTACATATGTTGAACGTTTCCTTGCTTCAGCTCTACGTAGGCATATGC
TCACGAGTAGCACAAGAACATACAAGAAAGATTGTACAACAAAAATCAAGCTTTGATAACGTTACTTAACCGCTGCCATTACCTACCA
CCAACTAGATTTCAGGCCTCAAAACACAATGCAAAACAAGTACAAAAGTTACCTAATATTGCTAGATCACATTACACGAGACCAAAC
GAAAAGATTTCTATGCTCTAGAAAAAGCTCTTCCCTACGCCTAAATATTTTCTAATTGTAGGTATGCTAACTTGCTTCTCTATAATTGTAAT
AATAATTAGCAATTGCAATAGGCTAGGGAGATTATCACAGTTACTTCTATTCTAAACACCATCTACGACTTGTATACGAGATAATTAGGTCT
GGAATCAGAAAGAAAAGGTAAATTAATAAGTCTCCTTGAAATGCAAGCGTGGAAGTACATCAATCTCACAGGAGCTTAGATGAACAC
GAAATAACAGATACCTCGCGGAGAGAAAATAGCCTTTCAAAACGAAAACGCGAGTAACTGCCCAAGACACTCTCCTGTCATAGCAAC
GATCTTCAATTTATAAATTCAAATCAATAATGCGGTGCTTGGTCATCAAAGAGAGGACTGGTGAAATTGCCGGACTCAAAGACTCATCCG
TCTTCCGCTCGTATGCACAATTATACACGGTATGATGGAGGCAGGAGAGGACAAAGAACGGGAGGGTAGGAGCTGGCTATCCCATGCA
GGCGTTATCAGGTGTCGCAATAGCAATACATGAAAGTCAGTACGGGGTACAAAAACCTTACCCGGAAAATGAAACAGCCTAATACTGT
CAACCGTTCCAGACCAGTCAACTGCAAGATTATTGTCAATTAGCAAATTAGATCAAGAAGTTTATCTCGACCGGAGCTCATTCAATTAGAT
TAGGTTAAGAGGTCTCGTCAACGGCAGCTTCCGAAGCGAGTCTGCCAGCTCCTCCGTACGGCTCAGTCAATTAGGACTCTTGTAGATGC
ACATACAATTTCTAGCAGAGCTTCTGAGAGCGAGCAGGTAAGTACTAACAAGCCAAGATGACAATGACAGCACAAGCAAGTACTCTTT
TAAGATTAGATTAAGGAGGGTCTGCCGTCCGCCTTAGTGAGCTCCCTCATAGGACTCGACACAACATAGATACTAACACACTTCTATCA
ATGCATCAAATAGGTATCTATAGAAAAGGGTACGCACGCAAATCACTCGGGTCATATACATAACAAAACGACCATTACATCAAAATAGA
CTATCGTACGGTTAACTGCATATTTCTTCGCCAACATTTGTACACGATATGACTTGGATTAACATGGGGGAGCGTTGGGAAAGGTTTTTCAT
CAACTCAGCACTCAAAAATACCTGAATAGTTGTGTGCTATTTTCTACAAGCTATAAATAGCCTAGCGAACTTGTTAGGAGACATCCAACG
ATTAAAATTTTGCAATTGCCTCGTCACAGAACAGCCAATGACTAAACATTGCTCTTTATCTCACC...

Gene 1
(exons/introns)

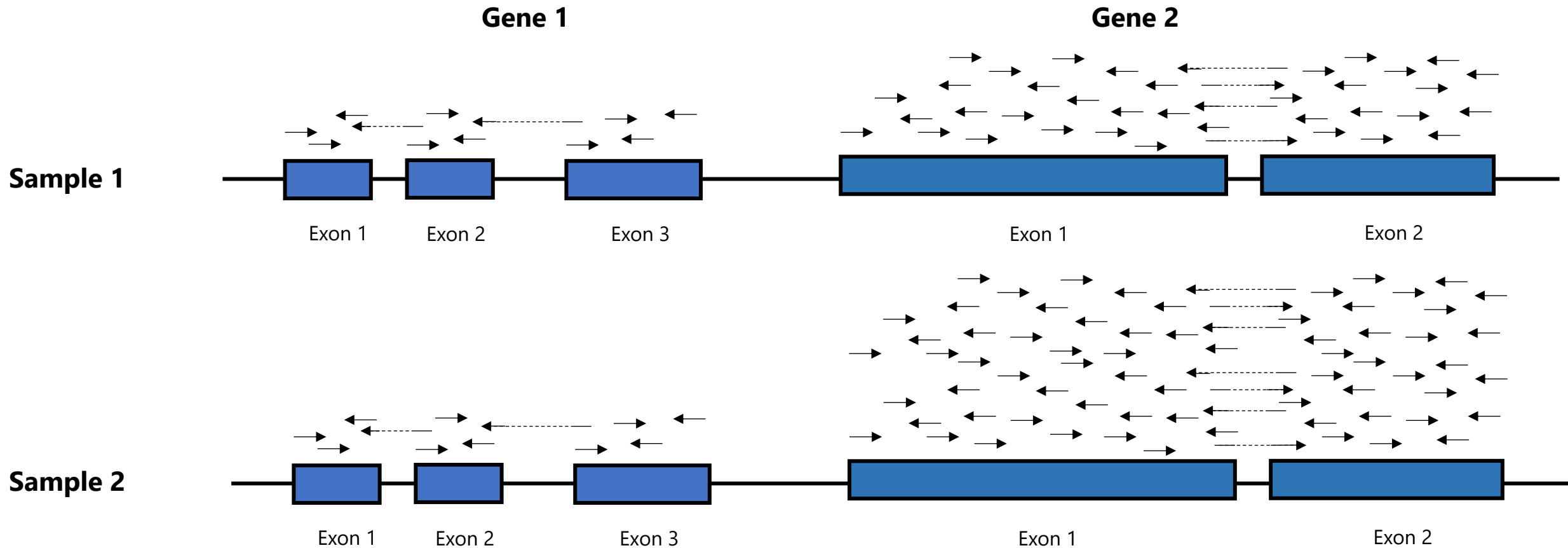
Dry lab - Read alignment to genome



Dry lab - Read alignment

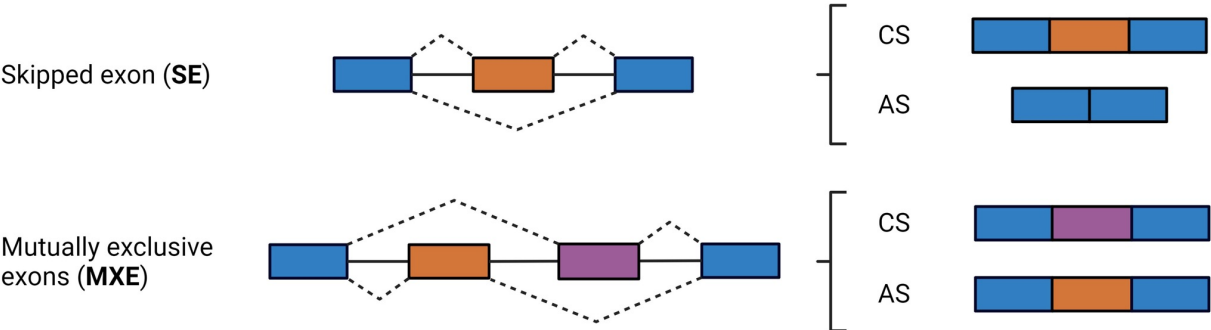
	Sample 1	Sample 2	...
Gene 1	500	650	...
Gene 2	2504	5896	...
...	...	...	...

Count matrix used to test
for differential expression
between samples



Dry lab - Reference transcriptome

Alternative
splicing



>Gene1.transcript1

```
GTTGCTTCTATTTAAGTTCTTTTTACTTGATATAAATATAGATAAATATATACAAAAGAAATAGTTCAGATAATCGCAAATAATTTTCAGAGC
GAACATTTAAAATGTGTTCTGATCCAAGTTTCACTGTTACTCCTCCTACATATGTTGAACGTTTCCTTGCTTCAGCTCTACGTAGGCATATGC
TCACGAGTAGCACAGAAGACATACAAGAAAGATTGTACAACAAAAATCAAGCTTTGATAACGTTACTTAACCGCTGCCATTACCTACCA
CCAAACTAGATTCAGGCCTCAAAACACAATGCAAAACAGGACTGGTGAAATTGCCGGACTCAAAGACTCATCCGTCTTCCGCTCGTATG
CACAATTATACACGGTATGATGGAGGCAGGAGAGGACAAAGAACGGGAGGGTAGGAGCTGGCTATCCCATGCAGGCGTTATCAGGTG
TCGCCAATAGCAATACATGAAAGTCAGTACGGGGTACAAAAACCTTACCCGGAAAATGAAACAGCCTAATACTGTCAACCGTTCCAGAC
CAGTCAACTGCAAGATTATTGTCAATTAGCAAATTAGATCAAGAAGTTTATCTCGAC
```

>Gene1.transcript2

```
GTTGCTTCTATTTAAGTTCTTTTTACTTGATATAAATATAGATAAATATATACAAAAGAAATAGTTCAGATAATCGCAAATAATTTTCAGAGC
GAACATTTAAAATGTGTTCTGATCCAAGTTTCACTGTTACTCCTCCTACATATGTTGAACGTTTCCTTGCTTCAGCTCTACGTAGGCATATGC
TCACGAGTAGCACAGAAGACATACAGTATGGAGGCAGGAGAGGACAAAGAACGGGAGGGTAGGAGCTGGCTATCCCATGCAGGCGTT
ATCAGGTGTCGCCAATAGCAATACATGAAAGTCAGTACGGGGTACAAAAACCTTACCCGGAAAATGAAACAGCCTAATACTGTCAACCG
TTCCAGACCAGTCAACTGCAAGATTATTGTCAATTAGCAAATTAGATCAAGAAGTTTATCTCGAC
```

>Gene2.transcript1

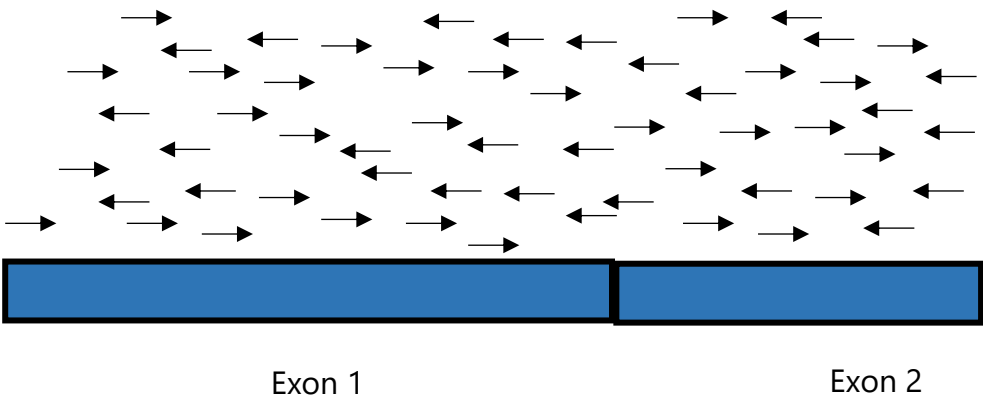
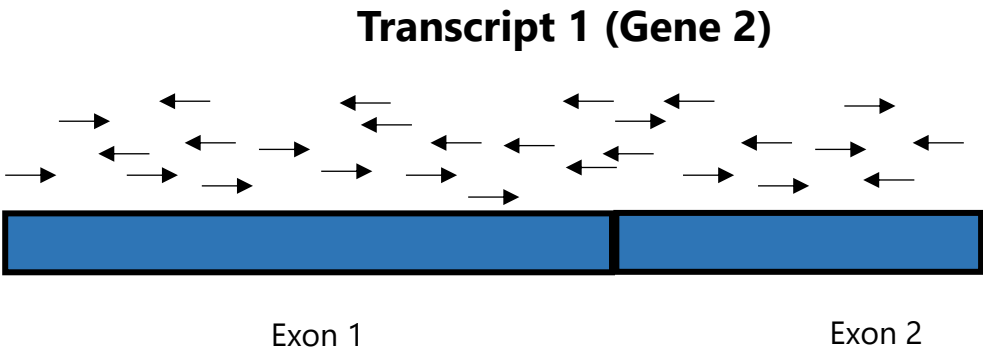
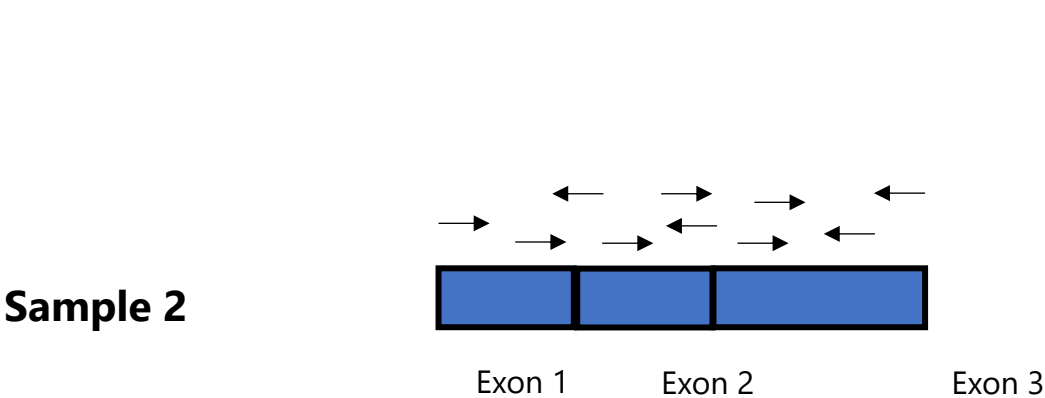
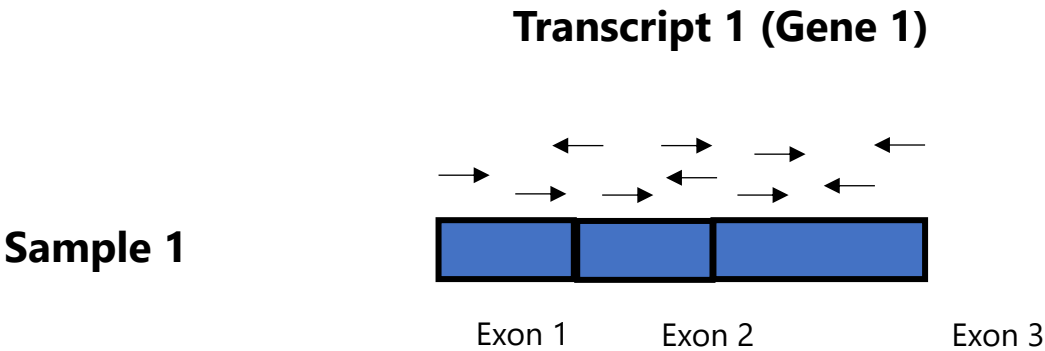
```
ATTTTCTACAAGCTATAAATAGCCTAGCGAACTTGTTAGGAGACATCCAACGATTAAAATTTTGCAATTGCCTCGTCACAGAACAGCCAA
TGACTAAACATTGCTCTTTATCTCACC...
```

Transcripts of gene 1
(2 alternatively spliced
isoforms)

Transcript of gene 2

Dry lab - Read alignment to transcriptome

	Sample 1	Sample 2	...
Gene1.Transcript1	416	650	...
Gene1.Transcript2	280	300	...
Gene2.Transcript1	2020	4589	...
...	...	...	...

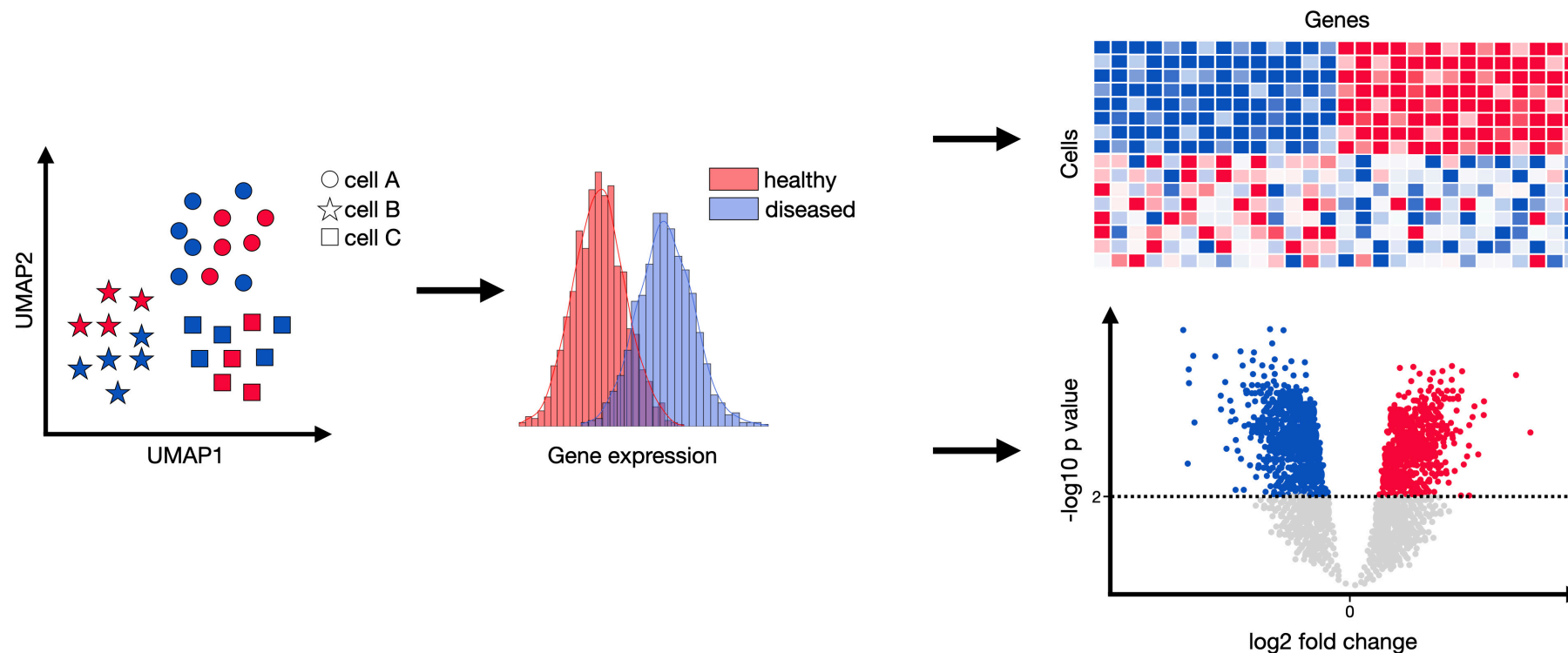


Recap questions

- What are the bioinformatic challenges with short read Illumina data?
- What is a read and how can it be used for expression estimation?
- How do alignment strategies differ between genomic and transcriptomic references?
- What are counts?

Dry lab – differential gene expression analysis

- Test if gene expression patterns (= abundances, counts) differ between sample types



Dry lab – differential gene expression analysis

- Test if gene expression patterns (= abundances, counts) differ between sample types

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
Gene 1	500	650	758	0	10	47
Gene 2	2504	5896	3587	10256	11589	9584
Gene 3	3587	5895	2598	257	894	685
Gene 4	2569	3512	2014	4512	3520	2597
...	...	...	...	...	...	...
Gene 20 145	78	95	63	150	26	34

1. Model counts for each gene:

Counts $\sim$ condition

Modelling count data

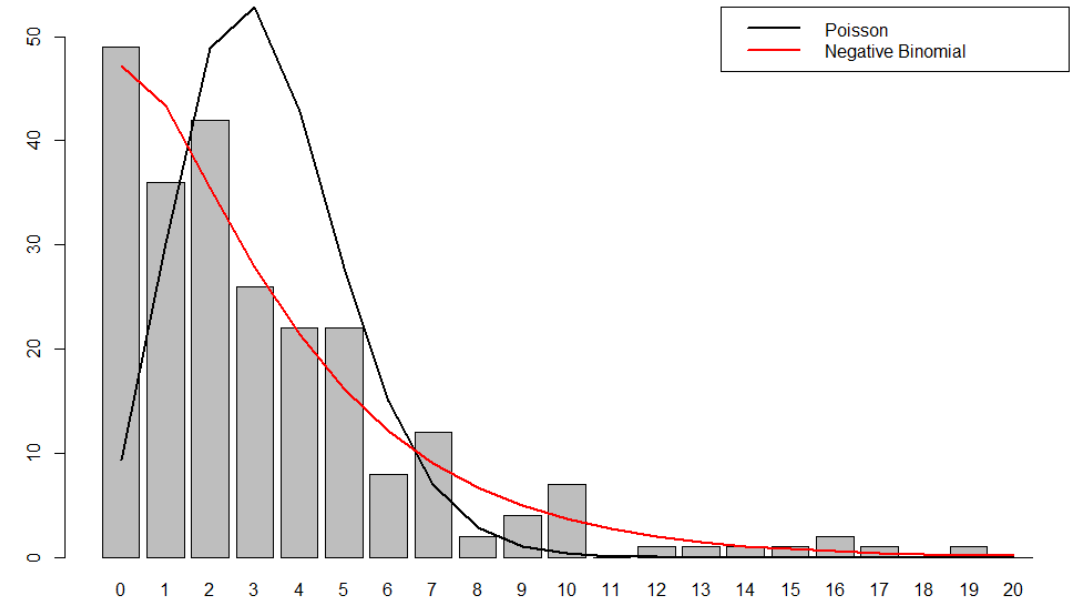
- Test if gene expression patterns (= abundances, counts) differ between sample types

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Counts $\sim$ condition

- Count data is not normally distributed (Poisson, Quasi-Poisson, Negative Binomial)

Gene counts



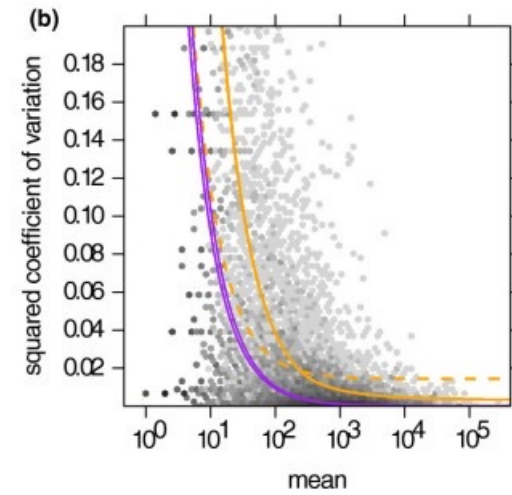
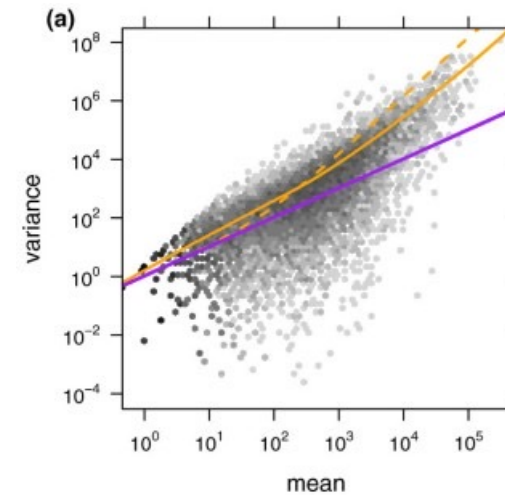
Modelling count data

- Test if gene expression patterns (= abundances, counts) differ between sample types

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Counts $\sim$ condition

- Count data is not normally distributed (Poisson, Quasi-Poisson, Negative Binomial)
- Variance depends on the mean (heteroscedasticity)



Modelling count data

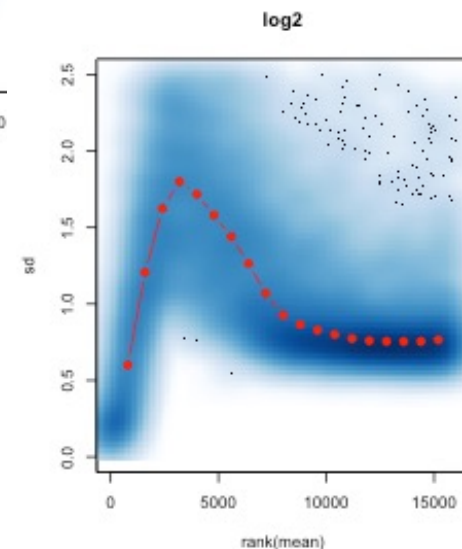
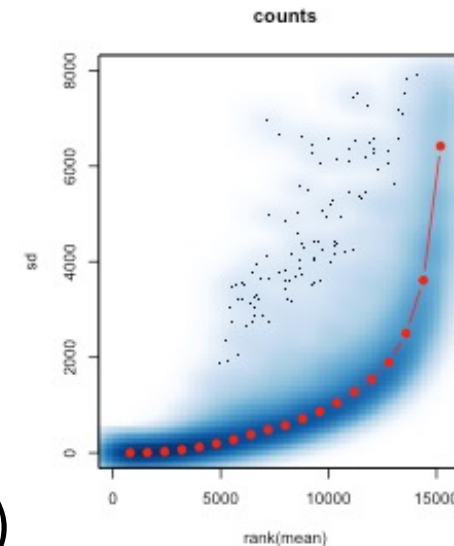
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1. Model counts for each gene:

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Solution: Fit negative binomial GLMs on (log2-)transformed counts



Modelling count data

- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene:

Counts $\sim$ condition

- Count data is not normally distributed
(Poisson, Quasi-Poisson, Negative Binomial)
- Variance depends on the mean (heteroscedasticity)

2. Test for differential expression: hypothesis tested that the LFC between groups $\neq 0$

Modelling count data

- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene (NB GLMs)
2. Test for differential expression

-> R packages **DESeq2** (Love et al., 2014) & **edgeR** (Chen et al., 2008) most widely used

Recap questions

- What means differential gene expression analysis?
- How is it performed?
- What is important to consider when analyzing/modelling count data?